Nicole Sandra-Yaffa Dumont

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Academic Appointments

Postdoctoral Researcher
Institute of Neuroinformatics (University of Zurich & ETH Zurich)
Dec 2025 – Present
Zürich, Switzerland

- **Supervisor:** Prof. Katharina Wilmes
- Awarded the **UZH Postdoc Grant**, beginning December 2026.
- **Predictive Coding (PC) & Models of Perception and Learning in Cortex**
  - Comparing prediction error modulation by learned uncertainty signals versus top-down attentional filtering
  - Building and comparing PC networks for multi-sensory integration

Education

Ph.D in Computer Science
Diploma in Theoretical Neuroscience
University of Waterloo
Sept 2019 – April 2025
Waterloo, Canada

- **Supervisors:** Prof. Chris Eliasmith and Prof. Jeff Orchard
- **Dissertation:** *Symbols, Dynamics, and Maps: A Neurosymbolic Approach to Spatial Cognition* [Link]
- **Neuro-Symbolic Modeling for Spatial Representation & Mapping**
  - Developed Spatial Semantic Pointers (SSPs) as a grid cell-inspired representation of continuous features, enabling the integration of symbolic structure and uncertainty in biologically plausible neural models.
  - Built spiking neural network models of path integration, cognitive mapping, and SLAM that use neurosymbolic methods to bind continuous and discrete features to create compositional embeddings.
- **Reinforcement Learning (RL) & Action Selection**
  - Designed biologically plausible online learning rules for RL in continuous-time settings using State Space Model-based memory systems.
  - Incorporated SSPs into Bayesian Optimization and deep RL frameworks, modeling predictive inference and decision-making under uncertainty.

Computational Mathematics (Masters of Mathematics – Co-op Program)
University of Waterloo
Sept 2017 – April 2019
Waterloo, Canada

Honours Mathematics and Physics (Bachelors of Science)
McMaster University
Sept 2012 – April 2017
Hamilton, Canada

Work Experience

Research Associate
Computational Neuroscience Research Group, University of Waterloo
April 2025 – Nov 2025
Waterloo, Canada

- Contributed to the incorporation of spatial reasoning and embodiment in Spaun 3.0, a large-scale functional brain model.

Neuromorphic Scientist

Applied Brain Research

Jan 2023 – Jan 2024
Waterloo, Canada

- Integrated a biologically-inspired SLAM algorithm into a MuJoCo simulation environment and deployed it on low-power neuromorphic hardware (Intel Loihi chip).

Research

h-index = 6, i10-index = 5, total citations = 165 (Google Scholar on 01/08/2026)

† = These authors contributed equally

Publications & Conference Proceedings

- Furlong, P.M.†, **Dumont, N.S.†**, Antonova, R.†, Orchard, J., & Eliasmith, C. (2026). Compositional neurosymbolic representations enable efficient active exploration. *Nature Communications*. https://doi.org/10.1038/s41467-026-75703-4 [Link]
- **Dumont, N.S.**, Furlong, P. M., Orchard, J., & Eliasmith, C. (2023). Exploiting semantic information in a spiking neural slam system. *Frontiers in Neuroscience*, *17* [Link]
- **Dumont, N.S.†**, Stöckel, A.†, Furlong, P. M.†, Bartlett, M., Eliasmith, C., & Stewart, T. C. (2023). Biologically-based computation: How neural details and dynamics are suited for implementing a variety of algorithms. *Brain Sciences*, *13*(2) [Link]
- Voelker, A. R., Blouw, P., Choo, X., **Dumont, N.S.**, Stewart, T. C., & Eliasmith, C. (2021). Simulating and predicting dynamical systems with spatial semantic pointers. *Neural Computation*, *33*(8) [Link]
- Coleman, T. F., **Dumont, N.S.**, Li, W., Liu, W., & Rubtsov, A. (2021). Optimal pricing of climate risk. *Computational Economics* [Link]

Conference Proceedings & Posters

- **Dumont, N.S.** & Wilmes, K. A. (2026). Multi-sensory integration in predictive processing [Accepted, poster presentation scheduled for Bernstein Conference, Sep 28 – Oct 1, 2026]
- Simone, K., Steffen, L. C., **Dumont, N.S.**, Yu, S., Ly, H., Damberger, G., & Eliasmith, C. (2026). Semantic map learning and externalization in an embodied neural agent: A comparison to human behavioral and neural data. *CogSci*, *48*(48)[Link]
- Furlong, P. M., Simone, K., **Dumont, N.S.**, Bartlett, M., Stewart, T. C., Orchard, J., & Eliasmith, C. (2024). Biologically-plausible markov chain monte carlo sampling from vector symbolic algebra-encoded distributions. *ICANN* [Link]
- Furlong, P. M., **Dumont, N.S.**, & Orchard, J. (2024). A recurrent dynamic model for efficient bayesian optimization. *NICE* [Link]
- Bartlett, M., Simone, K., **Dumont, N.S.**, Eliasmith, C., & Orchard, J. (2023). Improving reinforcement learning with biologically motivated continuous state representations. *ICCM* [Link]
- **Dumont, N.S.**, Orchard, J., & Eliasmith, C. (2022). A model of path integration that connects neural and symbolic representation. *CogSci*, *44*(44) [Link]
- Bartlett, M.†, **Dumont, N.S.†**, Furlong, M. P., & Stewart, T. C. (2022). Biologically-plausible memory for continuous time reinforcement learning. *ICCM* [Link]
- **Dumont, N.S.**, Stewart, T. C., & Eliasmith, C. (2021). Spiking neural network model of simultaneous localization and mapping with spatial semantic pointers [Poster presented at CoSyNe, February 23–26, 2021] [Link]
- **Dumont, N.S.** & Eliasmith, C. (2020). Accurate representation for spatial cognition using grid cells. *CogSci* [Link]

Manuscripts

- Sainz Villalba, L., Furlong, P. M., Bartlett, M., & **Dumont, N.S.** (2026). Neural representational geometry of feature binding operations. *bioRxiv* [Link]

Patents

- Voelker, A. R., Eliasmith, C. D., Blouw, P., Stewart, T., & **Dumont, N.S.** (pending, filed November 5, 2021). Methods and systems for simulating and predicting dynamical systems with vector-symbolic representations [US Patent App. 17/520,379] [Link]

Teaching and Supervision

- **Masters supervision (2026-present):** Supervising a masters student thesis project on learning compositional neural representation for planning and reasoning with tree diffusion.
- **Student supervision (2020):** Supervised an undergraduate research assistant during a four-month project developing a novel algorithm for learning successor features for reinforcement learning.
- **Introduction to Computational Mathematics (2022):** Sole instructor for two sections of this course, teaching over 120 students. Planned course content and developed lectures, assignments, and exams.
- **Logic and Computation (2020):** Served as an Instructional Apprentice for an introductory course in mathematical logic. Led tutorials, graded assignments, and provided assistance during office hours.
- **Teaching Assistantships (2017-2022):** Served as a TA in six undergraduate courses in mathematics and computer science.

Academic Service

- **Leadership & Organization**
  - Topic Area Leader, Spatiotemporal Dynamics in Neural Computation, Telluride Neuromorphic AI Workshop (2026)
  - Topic Area Leader, Language and Thought, Telluride Neuromorphic Cognition Engineering Workshop (2025): Successfully proposed and led a topic area; delivered tutorials on cognitive mapping models; mentored PhD-level researchers through the full project lifecycle from conceptualization to demonstrations for the public.
  - Invited Speaker & Co-Organizer, Telluride Neuromorphic Cognition Engineering Workshop (2024)
- **Instruction & Training**
  - Instructor, Nengo Summer School (2023, 2024, 2025): Led tutorials and mentored participants developing neural models using the Nengo framework.
- **Peer Review:** Reviewer for *Nature Communications*, *Communications Biology*, *CogSci 2025*, and *Frontiers in Computational Neuroscience*

Awards & Distinctions

- UZH Postdoc Grant (2026)
- Marie Skłodowska-Curie Postdoctoral Fellowship (MSCA-PF) Seal of Excellence (2026)
- Best New Neuromorph, Telluride Neuromorphic Cognition Engineering Workshop (2024)
- European Neural Network Society Best Paper Award (2024)
- Barbara Hayes-Roth Award for Women in Math and Computer Science (2022)
- GO-Bell Scholarship (2019–2021)
- Provost Doctoral Entrance Award for Women (2019)
- Keith & Debbie Geddes Graduate Scholarship (2019)
- University of Waterloo Graduate Scholarship (2018–2019)
- Emanuel Williams Scholarship in Physics (2014)